MADESWARAN K

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Executive Summary

Mechanical Engineering student with hands-on experience in CAD design, product development, and renewable energy projects. Skilled in SolidWorks, AutoCAD, ANSYS, and technical documentation, aligning with engineering and design-focused roles. Certified SOLIDWORKS CAD Design Associate (CSWA) with a strong ability to contribute to innovative engineering solutions and collaborative project environments.

Technical Skills

- **CAD & Design:** SolidWorks, AutoCAD (Basic), 3D Modeling, Assembly Design, Engineering Drawing
- **Analysis & Simulation:** ANSYS Workbench (Basic), ANSYS APDL (Basic)
- **Documentation & Productivity Tools:** MS Word, MS PowerPoint, Technical Documentation, Technical Report Preparation
- **Measurement & Inspection:** Vernier Caliper, Micrometer, Engineering Measurements, Basic GD&T Concepts
- **Professional Skills:** Problem Solving, Team Collaboration, Time Management, Leadership

Projects

Design and Fabrication of Tractor-Mounted Hydraulic Backhoe (2026) — Jan-May (2026)

- Developed a detachable tractor-mounted hydraulic backhoe that converts conventional tractors into multifunctional excavation machines for farming and land management applications. Implemented a universal chassis, hydraulic arm mechanism, and bucket system for efficient digging operations.
- Supported through EDII-TN Voucher A Grant Funding, with the product currently in the development phase (TRL 3) toward commercialization.
- Built using CAD Design, Hydraulic Systems, Mechanical Fabrication, Welding, Product Development, Assembly, and Testing.

Development of an Automated Robotic System for Urban Drainage Cleaning (2026) — Jan-May (2026)

- Developed a six-wheeled robotic drainage cleaning system with a rotating drill mechanism to clear blockages and a front-mounted filter unit to collect waste for easy disposal.
- Built using CAD Design, Robotics, DC Motors, Mechanical Fabrication, Circuit Design, and Prototype Testing.

Sustainable Power Generation Using Inline Hydro Turbines and Solar PV — Aug-Dec (2025)

- Developed a hybrid power generation system using inline hydro turbines and solar PV panels for sustainable energy production.
- Designed turbine blades in SOLIDWORKS and manufactured prototypes using 3D printing.
- Built using SOLIDWORKS, 3D Printing, Hydro Turbines, Solar PV Systems, Mechanical Fabrication, and Testing.

A Study on Wear Resistance of Al-Si Alloys Produced in Stir Casting — Jan-May (2025)

- Developed an Al-Si-Ti alloy using the stir casting process to improve wear resistance for automotive piston applications. Prepared material compositions, performed casting operations, and evaluated wear performance using a pin-on-disc test.
- Investigated the effect of titanium addition on the wear behavior and mechanical properties of Al-Si alloys through experimental testing and analysis.
- Built using Stir Casting, Material Characterization, Wear Testing, Metallurgical Analysis, Mechanical Testing, and Data Analysis.

Design and Manufacture of Simple Gear Train — Aug-Dec (2024)

- Developed and manufactured a simple gear train mechanism for efficient motion and torque transmission. Designed gear and shaft components, performed design calculations, and ensured smooth operation through assembly and testing.
- Manufactured gear train components using Wire Cutting, CNC Drilling, and Milling processes, including keyway machining and precision hole drilling for assembly.
- Built using SOLIDWORKS, Gear Design, Design Calculations, CNC Drilling, Wire Cutting, Milling Machine Operations, Assembly, and Testing.

Education

B.E Mechanical Engineering - K. Ramakrishnan College of Technology - Expected May 2027 -**CGPA - 8.16 out of 10**
HSC- Govt Syed Murthuza Higher Secondary School - Apr 2023 - **Percentage - 65**

Awards & Accomplishments

- Received EDII-TN Voucher A funding for the Tractor-Mounted Hydraulic Backhoe project.
- Advanced the project to TRL 3 (Technology Readiness Level 3).
- Successfully completed the SOLIDWORKS CAD Design Associate (CSWA) certification from Dassault Systèmes (Nov 2025).
- Participated in "The Catalyst 2026" event conducted by OnlyInterns at IITM Research Park, Chennai.